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Wireless Sensor Node(s) And Remote Data Receiver Using Arduino

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Wireless sensor node is a sensor node with a wireless transmitter. It has one or more sensing elements with amplifier and signal conditioning circuit or a digital/smart sensor, a microcontroller unit, modulator and transmitter with antenna, and a battery. Powered by a battery or solar power it senses temperature, sunlight, soil moisture, or tremor, etc and transmits the signal to a remote receiver.

Many such sensor nodes are placed at regular distances in a big geographical area to cover the whole area. All of them send their data to a receiver that is placed in a central data monitoring and storage system. All the sensor nodes transmit data (value) periodically. The receiver receives the data from all such nodes and stores and displays it for monitoring and controlling purpose.

Let us have a look at some examples.

In an agriculture field, the automatic irrigation system has many soil moisture sensor nodes that are spread over the entire farm. These periodically transmit soil moisture level to one central receiver, which is connected to monitoring and control system. If soil moisture level in one particular area goes below the threshold level, the water pump, solenoid valve, etc in that area is automatically turned on. Thus, required soil moisture level of the complete farm is maintained.

At the border between two nations, sensor nodes with PIR proximity sensors may be placed at regular distance to detect any motion at the border and send warning/alarm message to the controlling station.

For a green house, proper temperature, humidity, and sunlight needs to be maintained. So, it is equipped with many sensor nodes that sense temperature, humidity, and sunlight at different places in the green house and sends the data to central control room periodically. The central control room increases/decreases cooling, humidity, or light intensity as per the collected sensor data.

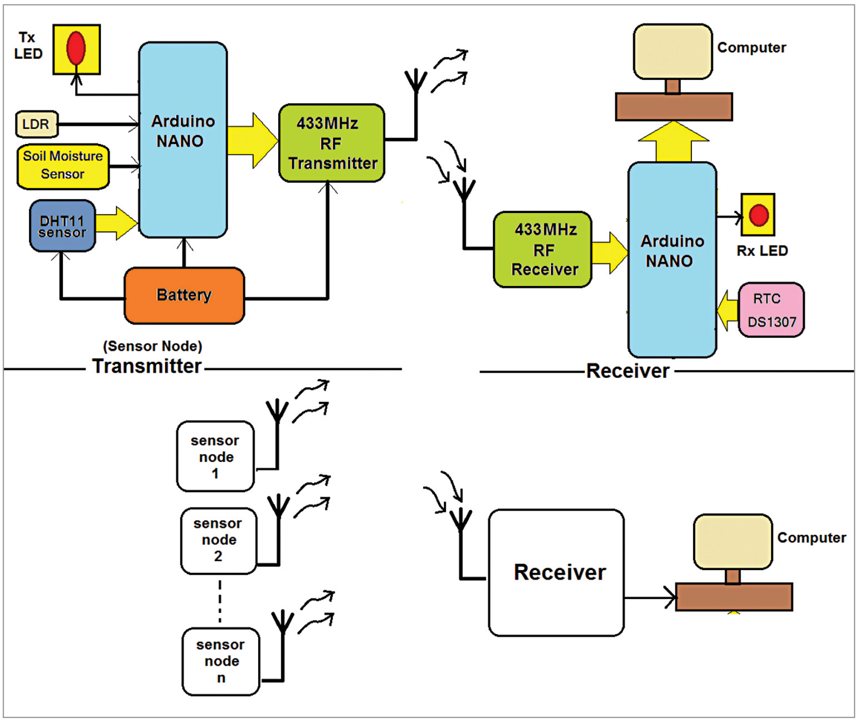

Fig. 1: Block diagram of the project

In this project the multiple sensors used are digital humidity and temperature sensor, light-dependent resistor (LDR) that senses ambient light, and soil moisture sensor that senses soil moisture content. The project also makes use of Arduino Nano as microcontroller (MCU) and 433MHz ASK RF transmitter module. There can be many such sensor nodes but only two such nodes are used here.

The receiver includes 433MHz ASK RF receiver module, real-time clock (RTC) module, and Arduino Nano. Both sensor nodes transmit data of sensed values of temperature, humidity, ambient light, and soil moisture content. The receiver receives the values from both sensors (total eight values) and gives them to computer, which displays these values and stores for future use.

As shown in block diagram of the project in Fig. 1, there are two distinct parts—transmitter (sensor node) and remote data receiver. The sensor node consists of different sensors, microcontroller, RF transmitter, and battery.

Sensors. DHT11 is a smart sensor that senses surrounding temperature and humidity and sends this data to microcontroller. LDR senses ambient light, and soil moisture sensor senses soil moisture content.

Microcontroller. Arduino Nano board is used as a microcontroller to read the data from all three sensors and transmit it using RF transmitter.